University
Mohammed VI
Polytechnic

College of
Computing

Deliverable 6: SQL – DDL and DML

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Team Name	Los primeros
Member 1	Sarah BOUCHMAL
Member 2	Douaa DYA
Member 3	Fatima Zahra BENJAA
Member 4	Douae CHARAFI

---

1 Introduction

This deliverable focuses on the implementation and use of the RLMS database through SQL. It covers the creation and population of relational tables, data manipulation, and information retrieval. The objective is to apply SQL concepts to manage the database and retrieve meaningful information while respecting the integrity requirements of the application.

2 Requirements

The main requirements addressed in this lab are:

- Implement the RLMS relational schema using SQL DDL.
- Define primary keys, foreign keys, and other appropriate constraints.
- Populate the database with representative data.
- Perform data manipulation operations, including insertions and updates.
- Retrieve information using SQL queries involving joins, aggregation, set operations, and nested queries.
- Identify integrity rules that cannot be enforced using basic SQL constraints.

3 Methodology

The implementation followed the relational schema developed in the previous labs. Tables were created with appropriate data types, primary keys, foreign keys, and `NOT NULL` constraints. The database was then populated with sample data representing the different entities and relationships of the RLMS application.

SQL queries were developed using the appropriate operations according to the information required. Assumptions were made where necessary to complete the implementation and populate the database.

4 Implementation & Results

The RLMS database was implemented and populated using MySQL Workbench. Data manipulation statements were executed to insert and update records. Queries were then executed to retrieve information from the database and verify the relationships between its tables.

The SQL scripts and execution logs provide the implementation details and the corresponding results.

Output | Result #1 | Result #2 | Result #3 | Result #4 | Result #5

pid	lastName	firstName	email
6	Alaoui	Karim	karim@um6p.ma
3	Amrani	Sara	sara@um6p.ma
2	Benjaa	Fatima Zahra	fatimazahra@um6p.ma
1	Bouchmal	Sarah	sarah@um6p.ma
10	Charafi	Douae	douae@um6p.ma
9	Chraibi	Imane	imane@um6p.ma
5	Dya	Douaa	douaa@um6p.ma
11	Kettani	Hiba	hiba@um6p.ma
4	Mansouri	Youssef	youssef@um6p.ma
13	nani	Hayat	Hayat@um6p.ma
8	Saidi	Mehdi	mehdi@um6p.ma
12	Tahiri	Anas	anas@um6p.ma
7	Tazi	Lina	lina@um6p.ma

Figure 1: result query 1

Output | Result #1 | Result #2

discipline
Physics
Chemistry
Engineering
Biology
Materials Science

Figure 2: result query 2

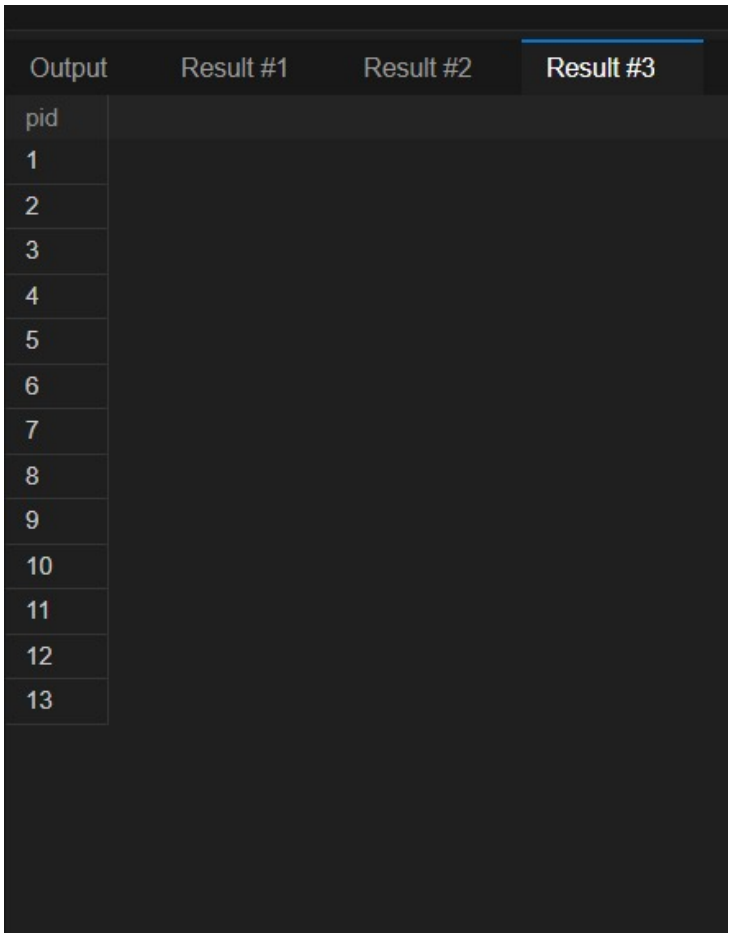


Figure 3: result query 3

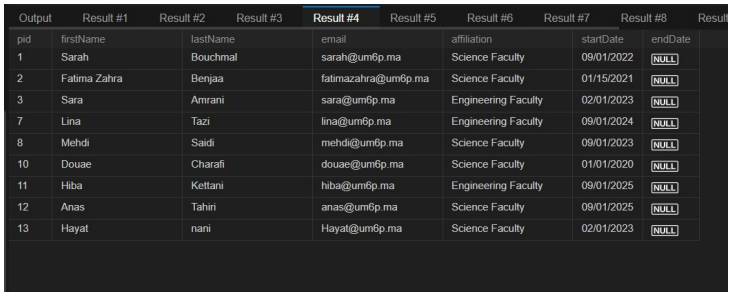


Figure 4: result query 4

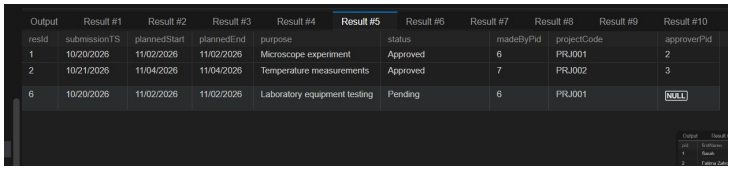


Figure 5: result query 5

Output | Result #1 | Result #2 | Result #3 | Result #4 | Result #5 | **Result #6**

firstName	lastName	projectTitle	role
Sarah	Bouchmal	Optics Research	PI
Fatima Zahra	Benjaa	Optics Research	Co-PI
Fatima Zahra	Benjaa	Thermal Analysis	PI
Sara	Amrani	Water Quality Study	PI
Karim	Alaoui	Optics Research	Collaborator
Lina	Tazi	Water Quality Study	Co-PI
Mehdi	Saidi	Smart Materials	Collaborator
Douae	Charafi	Smart Materials	PI
Hiba	Kettani	Thermal Analysis	Collaborator

Figure 6: result query 6

Result #7 | Result #8 | Result #9 | Result #10 | Result #11

budgetline	firstName	lastName
BUD001	Sarah	Bouchmal
BUD002	Fatima Zahra	Benjaa
BUD005	Fatima Zahra	Benjaa
BUD003	Sara	Amrani
BUD004	Douae	Charafi

Figure 7: result query 7

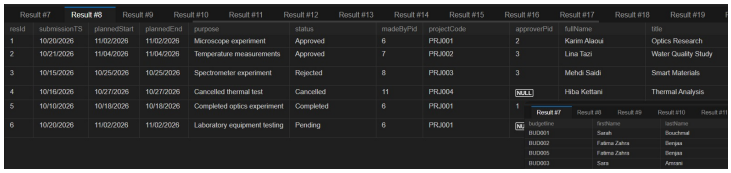


Figure 8: result query 8

Result #7 | Result #8 | **Result #9** | Result #10 | Result #11 | Result #12

resId	serialNo	commercialName	name
1	EQ001	OptiScope X100	Physics Laboratory
5	EQ001	OptiScope X100	Physics Laboratory
6	EQ001	OptiScope X100	Physics Laboratory
5	EQ005	OptiScope X100	Biology Laboratory
1	EQ002	ThermoPro T20	Physics Laboratory
2	EQ002	ThermoPro T20	Physics Laboratory
3	EQ003	SpectraScan 500	Chemistry Laboratory
4	EQ004	Centrifuge C300	Engineering Laboratory

Figure 9: result query 9

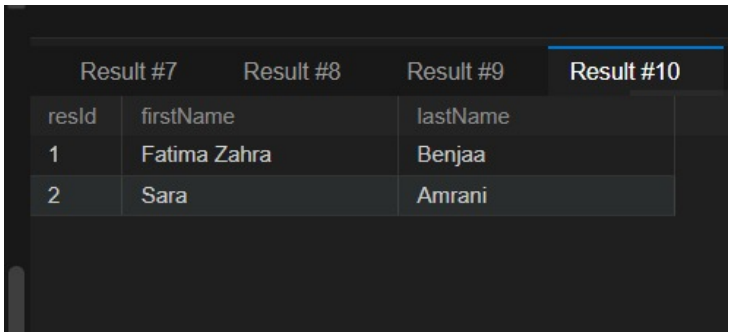


Figure 10: result query 10



Figure 11: result query 11



Figure 12: result query 12

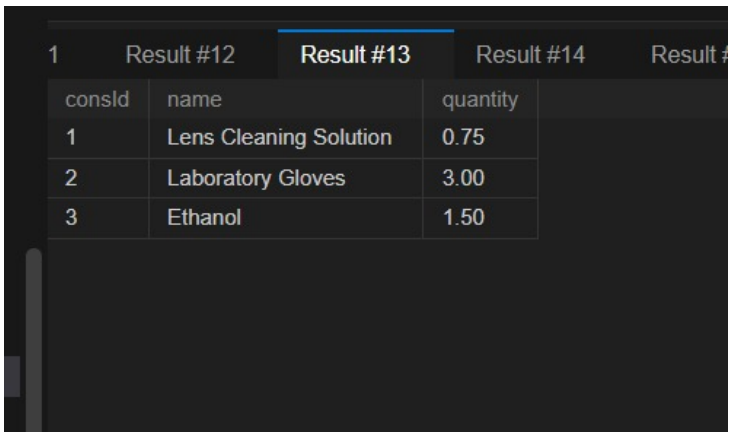


Figure 13: result query 13

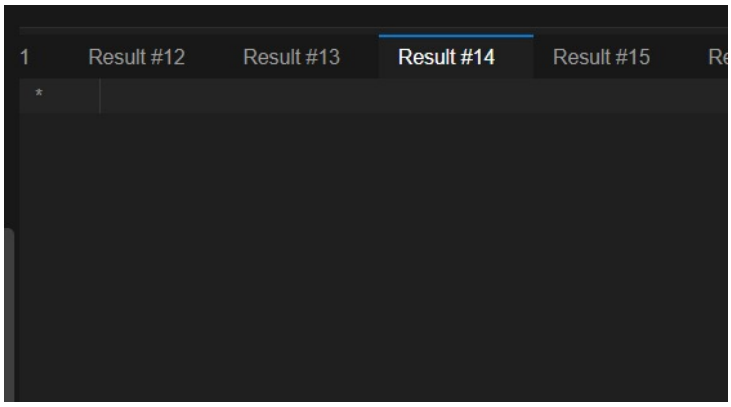


Figure 14: result query 14

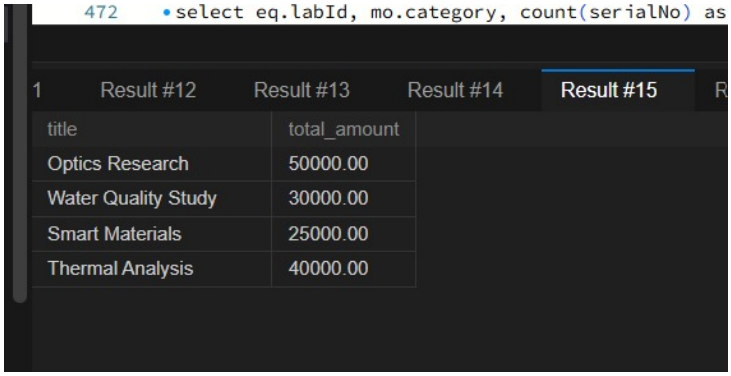


Figure 15: result query 15

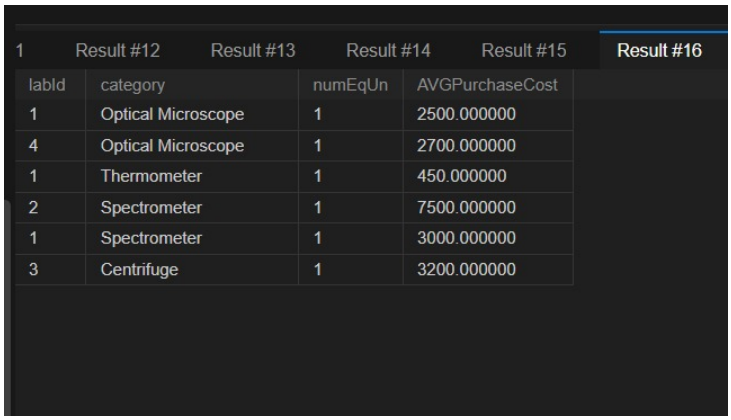


Figure 16: result query 16

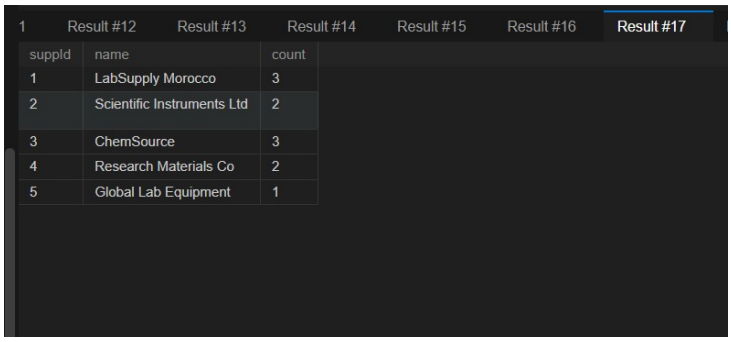


Figure 17: result query 17



Figure 18: result query 18



Figure 19: result query 19

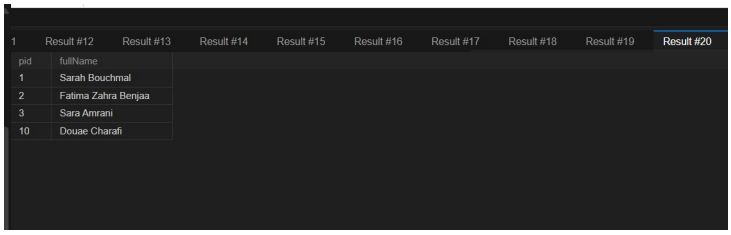


Figure 20: result query 20

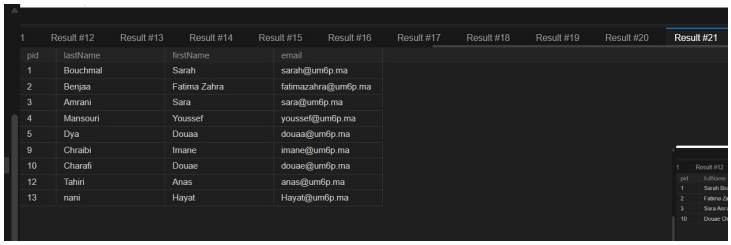


Figure 21: result query 21

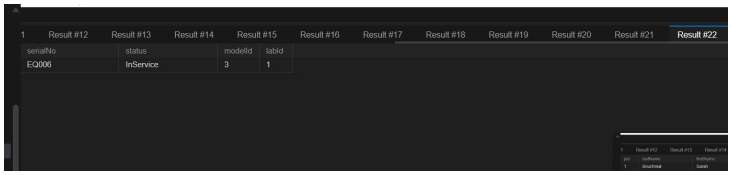


Figure 22: result query 22

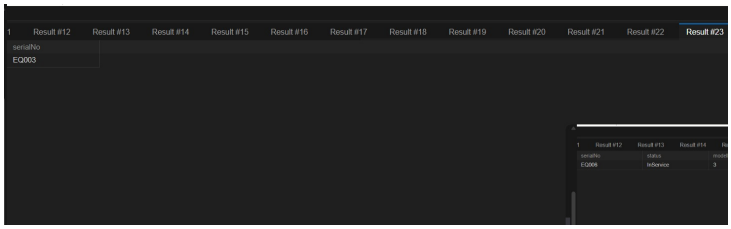


Figure 23: result query 23

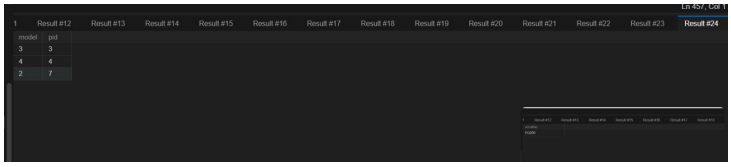


Figure 24: result query 24

5 Discussion

One of the main challenges was translating the relational schema into SQL tables while maintaining the correct relationships and integrity constraints. Another challenge was writing queries involving multiple tables, grouping, aggregation, and nested conditions.

The implementation also highlighted that some business rules cannot be enforced using basic SQL constraints and require more advanced mechanisms.

5.1 Part 4 – Open Question

Some RLMS integrity requirements cannot be enforced using only primary keys, foreign keys, and `NOT NULL` constraints:

- **Reservation eligibility:** Administrative staff must not be allowed to make reservations. The database must check the person's subtype when a reservation is created, which cannot be guaranteed by a foreign key alone.
- **RE1 – Every Reservation must reserve at least one EquipmentUnit.** A foreign key guarantees that every reservation resId in the Reserves table exists in the Reservation table. However, it does not guarantee that every resId in the Reservation table is in the Reserves table. The database needs to check that every reservation has at least one associated equipment unit, particularly after a reservation is inserted.
- **PE6. Each student is supervised by at least one academic staff member** A foreign key guarantees that every student in the Advises table exists in the Student table. However, it does not guarantee that every student in the Student table is in the Advises table. The database needs to check that every student has at least one academic staff who supervises them..

6 Conclusion

This lab provided practical experience in implementing and managing a relational database using SQL. It also demonstrated how SQL queries can retrieve information from related tables and highlighted the importance of advanced mechanisms for enforcing complex integrity rules.